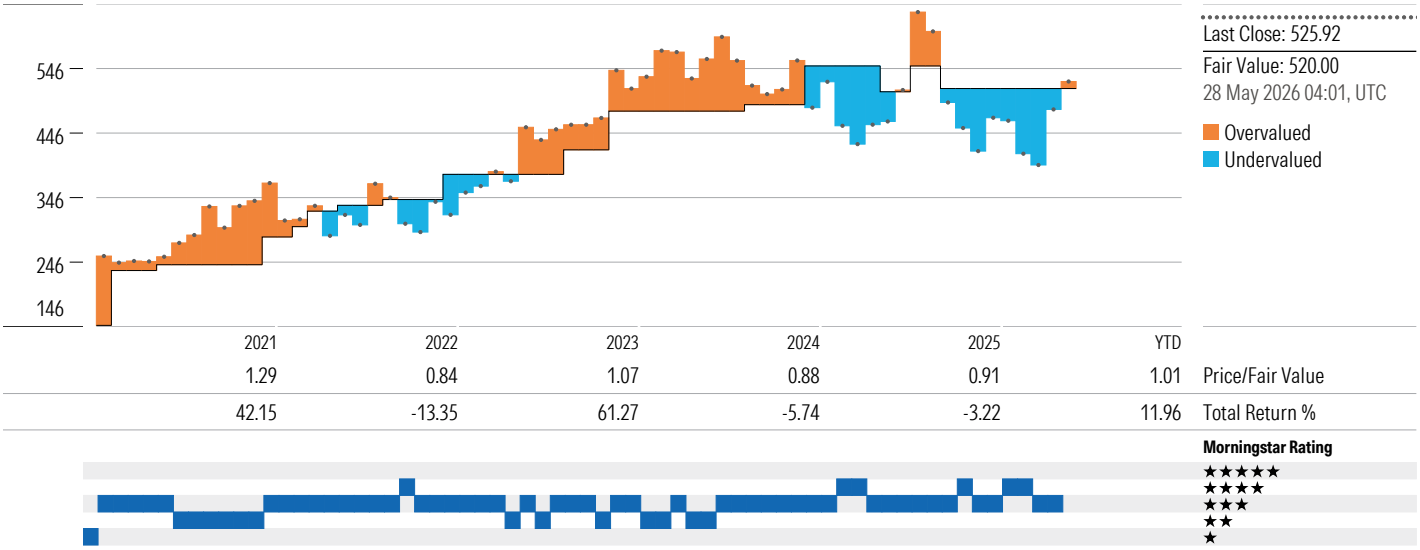
Synopsys Inc SNPS ★★★ 28 May 2026 04:03, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
525.92 USD 27 May 2026	520.00 USD 28 May 2026 04:01, UTC	1.01	100.75 USD Bil 27 May 2026	Wide	Large Blend	High	Exemplary	27 May 2026 05:00, UTC

Price vs. Fair Value



	2021	2022	2023	2024	2025	YTD	
	1.29	0.84	1.07	0.88	0.91	1.01	Price/Fair Value
	42.15	-13.35	61.27	-5.74	-3.22	11.96	Total Return %

Morningstar Rating

★★★★★
★★★★
★★★
★★
★

Total Return % as of 27 May 2026. Last Close as of 27 May 2026. Fair Value as of 28 May 2026 04:01, UTC.

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The primary analyst covering this company does not own its stock.

[1]The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

Synopsys Earnings: Solid Beat and Raise Quarter; Ansys Keeps Growing, Design IP Also Grows

Analyst Note Eric Compton, CFA, Director, 28 May 2026

Synopsys reported a solid second quarter, mildly beating our revenue and adjusted EPS expectations. Revenue grew 42% year over year, affected by the Ansys acquisition, and management raised its full-year outlook (EPS guide up 2% at the midpoint).

Why it matters: We thought the current quarter was a good result. In particular, the IP business is starting to show signs of life again, while Ansys growth continues to surpass expectations. Further details on the call also continued to support our AI pricing plus physical AI demand thesis.

- The IP business has been in turnaround mode, with efforts expected to start bearing fruit by year-end. This remains a key catalyst for Synopsys, and IP revenue was up 12% sequentially, the first sign of life since the second quarter of last year. We're hoping for double-digit annual growth next quarter.
- Ansys surpassed our expectations in the quarter, even excluding the accounting impact of gross revenue recognition. We had penciled in 10% annual growth previously, and it now looks like Ansys is growing closer to midteens, supporting our bullishness on products like multiphysics simulation.

The bottom line: We raise our fair value estimate to $520 from $515. For much of the year, we considered Synopsys shares undervalued, often trading below $450, but today they appear fairly valued. We still worry about potential China headwinds, but remain bullish on the rest of the business.

- Management commented on its partnership with Elliott Management, the appointment of Jesse Cohn

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Sector
Technology

Industry
Software - Infrastructure

Business Description

Synopsys is a provider of electronic design automation software and intellectual property products. EDA software automates and aids in the chip design process, enhancing design accuracy, productivity, and complexity in a full-flow end-to-end solution. Synopsys' comprehensive portfolio is benefiting from a convergence of semiconductor companies moving up the stack of technologies toward systems-like companies, and systems companies moving down-stack toward in-house chip design. The resulting expansion in EDA customers alongside secular digitalization of various end markets benefits EDA vendors like Synopsys.

to the board, and its focus on improving value capture and operational efficiency. To us, the key will be value capture, as IP and AI cause pricing model transitions.

- We liked the incremental details on agentic AI monetization, with over 20 customers evaluating over 25 specialized AI agents, with the pricing model set to subscription plus consumption. We expect incremental growth from pricing and value uplifts here, plus new value capture for IP products.

Business Strategy & Outlook Eric Compton, CFA, Director, 28 May 2026

Synopsys is a leading provider of software tools that engineers use to solve a fundamental industry challenge—designing complex semiconductor chips. It offers products that streamline chip building, enhance the efficiency of chip structure, automate processes, and minimize errors. Its solutions have become indispensable to virtually every major semiconductor company and a growing number of electronic systems companies globally. We see the primary growth drivers being when clients initiate new chip designs and price hikes to charge for additional value provided. We see a long runway for increased chip activity, driven by artificial intelligence-related chip development and increasing chip density and prevalence in electronic systems.

Synopsys has three segments. The electronic design automation, or EDA, business contributes half of the company's revenue. Synopsys is the leading digital EDA provider among peers like Cadence Design Systems and Siemens EDA (formerly Mentor Graphics). We foresee demand for EDA tools growing significantly as chip intricacy increases and the industry shifts toward multi-die or chiplet architectures, thus requiring more sophisticated design solutions, which Synopsys is well positioned to supply.

The intellectual property segment sells predesigned chip components. We observe IP complementing the EDA business and anticipate the industry's growing need for prebuilt IP blocks for standardized components to support IP demand.

Recently, Synopsys acquired Ansys, thus establishing the Simulation and Analysis segment. We like the strategic merger with Ansys as it positions Synopsys to expand its total addressable market to about $70 billion by 2030 while unlocking valuable synergies in simulation and verification. In addition, simulation plays a natural part in "Internet of Things," robotics, predictive maintenance and digital twins, in which Ansys is positioning itself to meet future demand.

We see a repositioning of the IP portfolio to focus on AI related needs, a more royalty heavy IP fee structure, AI enhanced EDA workflows, and a structural increase in demand for chips across the economy all as key drivers of growth for Synopsys.

Bulls Say Eric Compton, CFA, Director, 28 May 2026

- Synopsys' wide moat stems from high customer switching costs and intangible assets that virtually no

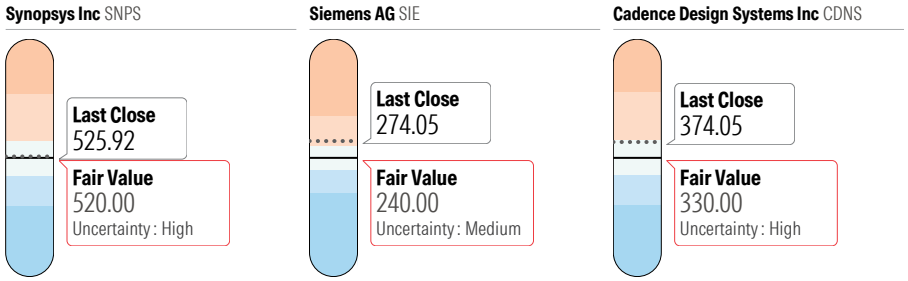
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Competitors



	Synopsys Inc SNPS	Siemens AG SIE	Cadence Design Systems Inc CDNS
Economic Moat	Wide	Wide	Wide
Currency	USD	EUR	USD
Fair Value	520.00 28 May 2026 04:01, UTC	240.00 27 May 2026 21:13, UTC	330.00 27 May 2026 16:29, UTC
1-Star Price	806.00	324.00	511.50
5-Star Price	312.00	168.00	198.00
Assessment	Fairly Valued 27 May 2026	Overvalued 27 May 2026	Fairly Valued 27 May 2026
Morningstar Rating	★★★28 May 2026 04:03, UTC	★★27 May 2026 05:00, UTC	★★★27 May 2026 05:00, UTC
Analyst	Eric Compton, Director	Matthew Donen, Director	Eric Compton, Director
Capital Allocation	Exemplary	Exemplary	Exemplary
Price/Fair Value	1.01	1.14	1.13
Price/Sales	11.11	2.71	18.49
Price/Book	3.31	3.25	15.72
Price/Earning	121.74	28.34	87.19
Dividend Yield	0.00%	1.95%	0.00%
Market Cap	100.75 Bil	209.03 Bil	103.17 Bil
52-Week Range	376.18—651.73	198.00—279.60	262.75—383.80
Investment Style	Large Blend	Large Blend	Large Growth

other software vendor or startup can match.

- Secular tailwinds will drive new chip design starts, raise R&D intensity, and open up new markets for semiconductor-focused EDA and systems design tools, leading to excess revenue growth for 10 or more years.
- Synopsys' margins should only improve, supported by improving IP margins, greater AI-driven wallet share, and internal AI/ML optimizations that enhance operational efficiency.

Bears Say Eric Compton, CFA, Director, 28 May 2026

- Slower growth in IP revenue and even lower margins could continue as the business transitions to more custom, chiplet-focused sales.
- The Ansys acquisition is complex, and may not produce the synergies or growth Synopsys is counting on. If Intel's foundry fails on 14A, it could be shut down, representing a material headwind to Synopsys.
- Geopolitical tensions with China and China's focus on supporting domestic companies could affect

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Synopsys' ability to do business in the region and lead to slower revenue growth.

Economic Moat Eric Compton, CFA, Director, 28 May 2026

We believe Synopsys warrants a wide moat rating as a result of switching costs and intangible assets associated with the firm's vast product suite. Synopsys provides leading-edge electronic design automation tools and intellectual property that are critical to the semiconductor chip design process. We believe that Synopsys enjoys high customer switching costs and intangible assets in its sticky EDA segment, as well as intangible assets associated with its IP offerings. In our opinion, this customer stickiness and profitable IP portfolio will drive excess returns on capital for Synopsys over the next 10-20 years. Fundamentally, we believe that as long as the semiconductor industry exists and is innovating, Synopsys will be a necessary part of that innovation, earning excess returns on invested capital.

Engineers use electronic design automation software to design and build semiconductor chips. This software is indispensable, functioning as the backbone for developing advanced microchips. Without EDA software, there would be no modern semiconductor industry. If a company is designing semiconductor chips, or its overall systems are heavily reliant on semiconductors (think modern automobiles), it is using EDA software.

The EDA industry is dominated by three key players that control over 90% of the market: Synopsys, Cadence, and Siemens EDA. Synopsys and Cadence are the most dominant market leaders, with 38% and 36% share, respectively, collectively capturing over two thirds of the industry. This dominance is reinforced by near 100% retention rates and an 80%-85% recurring revenue model, reflecting both the critical role that EDA tools play in chip development and the loyalty the top EDA firms have from their clients. These are essentially the strongest retention rates a software company can have, implying customer lifetimes well over 20 years. The primary reasons clients stop using the services of Synopsys is that they go out of business or are acquired. We believe the top EDA firms' market dominance (essentially a stable oligopoly), evidence of extremely high switching costs (retention rates near 100%), and position as indispensable tools for processes where firms cannot afford failure—where a bad chip design could put a younger firm out of business, or at the very least have a severely negative impact on a more mature firm—all support a wide moat rating based on switching costs and intangible assets.

Switching costs represent the primary moat source for Synopsys. Chip design is an intricate process involving thousands of hours dedicated to fine-tuning workflows, integrating tools, and aligning processes. Synopsys' tools are tightly integrated within its clients' workflows, and transitioning to an alternative could lead to significant implementation risks such as a drop in productivity and complex data migration challenges. In a rapidly moving technological industry, we believe that the costs of switching vendors far outweigh the small technological benefit the client would get in switching vendors. In addition, engineers invest considerable time in becoming proficient with specific tools, and

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changing vendors would mean more training, additional ramp-up time, disruption to ongoing projects, and delayed time to market.

Further, firms cannot afford mistakes in the design process, as they put future revenue and the future of the business at risk. Minor design flaws can result in significant disruptions and exacerbated losses down the line, making it necessary for these tools to be reliable and trustworthy. Clients would find it impractical to change providers without jeopardizing project deadlines and the quality of the final product.

In support of intangible assets, we point to the overall reputation of top EDA firms, like Synopsys. When a company is considering which EDA tools to use, there are only three reliable choices. We find it highly unlikely that a company would entrust its design processes to an unknown, unproven upstart. It is simply not worth the risk. This hesitancy toward new vendors, along with gradual industry consolidation over time, has led to minimal investment in new alternatives. This is in stark contrast to other parts of the semiconductor industry, such as chip design (the clients of EDA firms), where new capital and companies jumped at the opportunity presented by new waves of demand, such as designing AI-optimized chips.

The barriers to entry in the EDA business are substantial, requiring substantial upfront capital, deep technical expertise, and cumulative industry experience that cannot be easily replicated. Synopsys and Cadence's continuous acquisitions to build their software portfolios have led to market consolidation and a robust pace of innovation, which has put them leagues ahead of smaller competitors. We observe that if either lacks a feature or process, they have both the resources and technical know-how to catch up and maintain their competitive advantage. A major advantage that Synopsys has over smaller competitors is its ability to offer an end-to-end design flow—a full suite of tools needed to design a chip from start to finish—thus reducing the appeal of small competitors that have fragmented solutions.

Another industry feature that gives us confidence in the wide moat rating is the fact that EDA is not cyclical. During uncertain times and economic downturns in the semiconductor market, firms can cut budgets, but one area they are loath to reduce is their core R&D budget. Innovation is essential to maintaining a competitive edge in the semiconductor industry. If a company starts cutting its EDA budget significantly, it means it is effectively firing its engineers and is likely facing serious financial difficulty—potentially even on the verge of going out of business.

Over the long term, we expect demand for EDA to only increase. As chips get smaller and more complex, and therefore more expensive to design, customers are willing to pay a premium for the latest tools that Synopsys has to offer. Increasing complexity and increasing difficulties in chasing improved performance also increase the value of the solutions EDA firms offer. To meet this demand, Synopsys has some of the highest R&D spending in the software space to ensure it remains at the cutting edge.

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Another factor increasing our confidence in the long-term demand outlook is the increasing diversity of Synopsys' client base, extending beyond semiconductor-focused companies to include what are called system-level customers. Fifteen years ago, nearly all of Synopsys' clients were semiconductor companies, whereas today, close to 45% are companies that produce the end products rather than the chips alone. For example, an automotive firm that has to design a car that uses many different semiconductor components will design the car with the overall system, including the semiconductors, in mind. It can use EDA software for this system-level design process. With products only becoming more complex and more electronic over time, we expect this demand trend will remain in place for decades to come. With the EDA industry consolidated around a few players, Synopsys' brand and unique technological advantage position it to capture pricing uplifts, as well as volume increase, from a broadening customer base, along with industry-leading margins, all of which ultimately flow into better returns on capital.

Synopsys' other primary business is its IP segment. This is becoming an increasingly important part of Synopsys. As of fiscal 2023, IP growth has outpaced Synopsys' core EDA growth, representing around 30% of the company's revenue.

Synopsys' IP solutions can be compared with LEGO building blocks. When designing a chip, it is helpful for engineers to use prebuilt blocks for parts of the chip that are common and undifferentiated, allowing designers to focus on customizing the differentiated aspects of their creation. For example, part of the chip may have to deal with interacting with a USB connection. Building out this part of the chip each time a new chip is designed is not an efficient use of engineering resources. Companies like Synopsys will prebuild this part of the chip, and then sell that part of the design as IP. This allows clients to purchase these modular, undifferentiated parts of a chip's design for a fraction of what they would cost to develop in-house with their own engineering team. Synopsys can sell this IP at an attractive price because it is able to depreciate its cost of development over a large base of clients.

We believe this business benefits from an intangible asset moat source, driven by the breadth of its IP library and Synopsys' unique position at the center of the semiconductor design workflow.

Synopsys has steadily expanded its IP library (through acquisitions and internal development), building a robust repository of standardized and custom solutions that differentiate it from smaller competitors. The extensive library, combined with continuous innovation in sectors like automotive and AI-focused applications, keeps Synopsys at the forefront of client demands and the pace of technological developments. Nearly half of Synopsys' workforce is dedicated to building and maintaining this library, a testament to the company's commitment to quality and breadth. This breadth makes Synopsys a natural first place to look for IP solutions as a designer.

The second key ingredient to Synopsys' moat in IP is its unique position at the center of semiconductor

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design. In theory, anyone could create many of these IP blocks. However, it is fundamentally more efficient at the industry level to have one or two firms do it and then sell those standardized design blocks to the rest of the industry, as opposed to having each design firm rebuild what are functionally the same design blocks for every project. This allows a firm like Synopsys—which is naturally already at the center of the design process for the industry with its software—to have a natural distribution advantage. This further allows Synopsys to depreciate the cost of development for each IP block across a wider base of customers. We hesitate to award the firm a cost advantage for this because, in theory, it does not cost Synopsys any less to develop these IP blocks, but the ability to sell them across a wider base of customers does allow Synopsys to sell them at a price that would be unpractical for a firm with too few customers, and at a price which is cheaper than what it would cost any of its customers to develop these IP blocks in house.

Synopsys has increased its market share in the IP business to over the last decade by taking market share from Arm, while also outpacing Cadence. The consistent quality of Synopsys' IP fosters long-term partnerships and deepens client loyalty. Furthermore, client-specific customized solutions developed with Synopsys make switching to other providers challenging, as it risks compromising design integrity and reliability.

Lastly, the type of IP portfolio matters. Arm has the most dominant market share, but it is focused only on processor IP. Meanwhile, Synopsys and Cadence concentrate on non-processor IP, which is growing fast; we forecast it will soon account for more than half of the IP market. With a fast-growing IP margin profile, we believe Synopsys' IP business will be a significant contributor to strong and improving ROICs and future growth.

Fair Value and Profit Drivers Eric Compton, CFA, Director, 28 May 2026

Our fair value estimate of $520 per share implies a fiscal 2026 adjusted price/earnings multiple of 35 times.

Over the next five (10) years, we forecast Synopsys' EDA segment revenue to grow at a 11% (10%) CAGR and the IP segment at a 10% (11%) CAGR. Overall, we believe Synopsys will grow at a 10% (10%) CAGR from 2026 through 2030 (2035), driven by AI-related demand, and above the historical industry growth rate of mid to high single digits. We believe this above average growth is warranted by several key trends, including AI demand, increasing chip complexity, the increasing proliferation of EDA tools to "systems" companies, the physical AI buildout, and ability to monetize AI tooling.

We are seeing generative AI demand leading to more chip design starts and R&D investment, fueling growth in EDA tools. Growing chip design complexity and the slowdown of Moore's Law have led to a shift to multi-die architectures, necessitating more sophisticated EDA solutions and increased use of EDA tools. We also expect rapid adoption of silicon content, that is, more chip demand from systems

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companies across various industries like automotive, high-tech, industrial, and aerospace, thus increasing adoption of Synopsys' EDA products, with increased robotics adoption another potential large tailwind.

We view the strategic acquisition of Ansys as expanding the total addressable market in these same systems companies. We think substantial growth will stem from the firm's competitive advantage in simulation and the ongoing rise in global connected devices and 5G rollouts, and Ansys will benefit from digital twins and predictive maintenance as these trends gain more popularity. We also expect Ansys to see increased annual contract value from increasing partnerships with CAD providers that are meeting new needs--such as visualization of simulation.

We also see growth opportunities in IP, as companies need the reusable design blocks—not only for prioritizing differentiated design elements while outsourcing the basic design work through IP—but also for accelerating design cycles and go-to-market strategies.

We expect GAAP (adjusted) operating margins to continually increase, from 13% (29%) in 2025 to 35% (40%) by 2035. This will be driven by strong growth and repositioning in the IP business (where we expect margins to improve as the business scales), AI-driven efficiencies, pricing opportunities through selling higher-value-added products (value-add increases driven by increased demand along with product innovation, like AI enhancements), and more typical scale-based efficiencies (software companies do not need to double the salesforce or general and administrative expenses to double sales, for example). Management targets adjusted operating margins in the mid-40s.

A stable business model and low cyclicality in revenue thanks to resilient demand during economic downturns gives us confidence in assigning a 7.5% cost of equity.

Risk and Uncertainty Eric Compton, CFA, Director, 28 May 2026

We assign Synopsys a High Morningstar Uncertainty Rating. This is primarily based on the uncertainty of future revenue growth. We expect EDA firms to see above-industry growth for the next two decades, driven by key structural tailwinds. The degree of uncertainty around this revenue upside and its implications for our fair value estimate lead to the High Uncertainty Rating.

We also assign the firm a cost of equity of 7.5%, the lowest discount rate we use. We do this because we see low risk to the overall business given its entrenched position, immunity to cyclicality, and strong financial position and margins. We view the business as low risk and high quality, but there is significant uncertainty around our fair value based on future revenue growth relative to the overall semiconductor market.

Key risks for Synopsys include headwinds in China, customer concentration, and operational execution risks related to the Ansys deal and future innovation.

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MORNINGSTAR®

Synopsys Inc SNPS ★★★ 28 May 2026 04:03, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
525.92 USD 27 May 2026	520.00 USD 28 May 2026 04:01, UTC	1.01	100.75 USD Bil 27 May 2026	Wide	Large Blend	High	Exemplary	27 May 2026 05:00, UTC

While previously roughly 15% of total revenue came from China, the company is now trending below 10%. New regulations on exports to China, along with increasing competition in the domestic semiconductor industry (China's long-term vision is for the Chinese market to consist of Chinese firms), could all pose headwinds to revenue growth.

Synopsys depends on a relatively small number of customers for large chunks of its revenue base. However, although it derives most of its sales from chip designers, it has been diversifying into systems companies. Synopsys garners approximately 10% of revenue from Intel alone, so losing a major client could affect the company.

Synopsys also faces execution and integration risks from the acquisition of Ansys and in the turnaround of its IP business, as well as future innovation risks in general, including the need to innovate to stay competitive with key rival Cadence and to take share in new addressable markets.

We believe Synopsys faces low levels of environmental, social, and governance risk.

Capital Allocation Eric Compton, CFA, Director, 26 Feb 2026

We assign Synopsys an Exemplary Capital Allocation Rating based on a sound balance sheet, exemplary investments regarding strategic priorities and execution, and appropriate shareholder distribution policies.

With solid profitability, low debt, and reasonable cash balances, we view the balance sheet as sound. With regards to shareholder distributions, Synopsys prioritizes capabilities-focused mergers and acquisitions to further its overall competitive positioning and growth first, and then uses excess capital for reinvesting into the business and share repurchases. Share repurchases primarily offset existing share issuance. The company does not pay a dividend, although perhaps at some point in the future it might. We view all of this as reasonable.

Regarding investments, we view the company's moves as exemplary. Synopsys was founded by Aart de Geus in 1986 to provide EDA solutions in a rapidly burgeoning chip design market. In January 2024, COO Sassine Ghazi took over the CEO role from de Geus, who moved to the executive chair role.

Over the years, Synopsys has engaged in numerous small tuck-in acquisitions and has made some sizable investments to emerge as a leader in the industry. We view these investments as exemplary because they have enabled Synopsys to keep up with industry changes, build out a complete product portfolio, and also add on additional business lines such as IP. All of these have increased shareholder value, and we think the company will continue to make such moves.

In July 2025, Synopsys acquired Ansys for approximately $35 billion in enterprise value. We think the Ansys acquisition is a sound deal for Synopsys, and once again we think Synopsys is making the right strategic move, building out the simulation portion of its product portfolio, aiding its existing customers in "silicon to systems" solutions, and playing on the increasingly important theme of "systems"

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companies. The deal expands Synopsys' addressable market and provides diversification outside of semis into companies doing systems-level design.

One area that has not worked out as well has been Synopsys' venture into security testing and software integrity through acquisitions like Coverity in 2014 and Black Duck in 2017. These businesses remained low-margin, and Synopsys' decision to sell aligns with a focused capital-allocation strategy, enabling the firm to concentrate resources on high-margin and high-growth areas. Another test for management will getting real momentum in the turnaround of the IP business by the end of 2026.

Analyst Notes Archive

Synopsys Earnings: Amid Small EPS Beat, Ansys Is Trending Well and The Dust Is Starting to Settle
Eric Compton, CFA,Director,11 Dec 2025

Synopsys reported solid Q4 results, with revenue growing 39% year over year, although ex-Ansys growth was closer to -3%. Adjusted EPS of $2.90 was a 4% beat compared with guidance. Why it matters: Overall, we were pleased with the results. Ansys' growth trended well, and 2026 guidance was largely positive, which should start to help reduce uncertainty. Ansys Q4 revenue of $668 million was roughly in line with our expectations, and the 2026 outlook of $2.9 billion was just ahead. The guide and management commentary confirm the business is growing at double-digit rates, which we sensed investors were concerned about pre-earnings. We viewed the 2026 outlook positively, as it confirmed that non-GAAP operating margins should start expanding materially again in 2026, exceeding 40%, and it set boundaries on where IP growth is likely to settle (we think low single digits in 2026), along with expenses. The bottom line: We are maintaining our fair value estimate of $515 and view shares as moderately undervalued. We value shares at a forward P/E of 36 (compared with a historical median of 35), although we expect that to be sub-33 times by 2026 as we expect EPS to grow 17% in 2027. We believe commentary on IP was largely positive, with management highlighting key product wins in AI-related products (like next-gen high-speed interconnects PCIe and UCIe and next-gen memory interface IP) while hinting that they had high certainty about 2026 roadmaps. We expect that combined Ansys + Synopsys product launches, along with AI-enhanced workflows, and even GPU-enhanced workflows (as discussed in the recent Synopsys + Nvidia partnership), will provide opportunities for additional price increase opportunities in 2026 and beyond.

Synopsys Model Update: We Have Lowered Our Fair Value Estimate Slightly After Additional Updates Eric Compton, CFA,Director,15 Sep 2025

Following third-quarter earnings, the first with Ansys, there were a number of moving parts and we have gone through a more extensive model update after our initial reaction. Why it matters: Our initial reaction to earnings had us lower our fair value estimate from $550 to $520, and after additional updates to our model we are lowering our fair value to $515. For our latest updates we reevaluated what

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the combined results of Synopsys and Ansys would look like, our assumptions around China-related revenue, and the total cost assumed for the acquisition. The bottom line: While we updated many of our underlying assumptions, we still view shares as undervalued, trading at $425 compared with our updated $515 valuation, a 4-star-rated 17% discount. We believe we have de-risked revenue assumptions related to China given what we view as the long-term direction of a gradual decoupling. We assume China-related revenue falls to almost $0 over the next decade. We reevaluated what the combined Synopsys/Ansys results will be. Our updated forecasts assume the combined entity will eventually reach gross margins of roughly 85% (Synopsys low 80s, Ansys close to 90%) and GAAP/adjusted operating margins of low 30s-low 40s. Big picture: This puts Synopsys slightly below management's long-term operating margin target of mid-40s. We still have the core (ex-China) business growing at a 10% CAGR over the next decade (in line with management's double-digit long-term target). We assume the IP business begins to recover by fiscal 2027, resuming its 9%-13% annual growth trajectory (versus midteens target), while margins recover to the low 30s, and the design automation segment maintains mid-40s margins. This puts our fair value at a normalized forward P/E ratio in the mid-30s by 2026, in line with the previous five-year average of 35. Our thesis depends on a recovery in IP, successful execution on Ansys, and no fundamental changes to the EDA industry's long-term structural tailwinds.

We Are Placing Synopsys Under Review Eric Compton, CFA,Director,11 Sep 2025

We are placing Synopsys under review to allow for further analysis and a model update. We plan to provide an update as soon as possible.-

Synopsys Earnings: IP Slowdown Causes a Dent and China Headwinds Drive Softer Outlook Eric Compton, CFA,Director,10 Sep 2025

Synopsys' third-quarter revenue grew 14% year over year, and non-GAAP operating margins came in at 38.5%, both falling short of FactSet consensus expectations. This was driven by an 8% year-over-year decline in IP revenue, exacerbated by a sluggish China recovery. Why it matters: A multitude of factors led to the IP revenue downturn—mainly the headwinds from the export restrictions to China, missteps in resource allocation, and heavy investments made for a major foundry customer that did not materialize, which we assume to be Intel. Even after the restrictions were reversed, Chinese customers remained hesitant to commit to multiyear deals, reflecting extreme purchasing caution. As such, the impact of export restrictions to China was felt even after the restrictions were reversed. While IP revenue can be lumpy quarter to quarter, we estimate all the above factors create a tough environment for IP to achieve the earlier targeted mid-teens growth rate for the segment. Management stated they are de-risking the outlook and expect a more transitional year for IP. The bottom line: We lower our fair value estimate for wide-moat Synopsys to $520 from $550, based on tougher near-term IP trends and pressure on China revenue. We felt shares were 12% overvalued pre-earnings, and with shares down

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25% after hours, we now see the stock as moderately undervalued. We think management's limited commentary on Ansys potentially spooked investors. There was limited information around Ansys' contribution, the synergy roadmap, or the cadence for next quarter. We still think the acquisition will be a success, but we admit that messaging could have been better. While we don't expect overall IP revenue to bounce back for a few more quarters, based on management's commentary around it being a time of transition, we still generally believe that chiplet-based IP will be valuable. Eventually, this business line will return to strength.

Synopsys and Cadence Design: EDA Firms Resume China Operations Following Trade Progress Eric Compton, CFA,Director,8 Jul 2025

Leading electronic design automation companies Synopsys and Cadence Design have received approval from the US Department of Commerce to recommence their operations in China. Why it matters: China accounts for nearly 11% of revenue for each company. A month ago, the EDA companies were ordered to halt China operations indefinitely, which meant a substantial corporate revenue cut as business from China would have effectively been cut to $0. The news that both firms can now resume operations essentially brings them back on their original trajectory of flat year-on-year China growth for Cadence, and below corporate average growth for the China business for Synopsys. We believe this reversal comes on the back of the broader trade deal that the US and China are working toward. The bottom line: Based on this development, we update our fair value estimate for Synopsys to $550, up from $510; and $297 for Cadence, up from $275. We incorporates Cadence and Synopsys' original guidance for the coming quarter and fiscal year, as we no longer assume business from China drops to zero. We also take into account the loss of revenue for both the EDA firms during the China export restriction, specifically from the time both companies received letters to halt China operations (May 29) till the time they were able to restart business in China (July 2). We believe that the one-month interruption shaves a few basis points off this year's run-rate, but it does not alter our fundamental view on the companies or the EDA industry. Coming up: For Synopsys, the retraction of the China export restriction improves the prospects of Chinese regulatory approval for the merger with Ansys. Comparing our updated fair value estimate with the market price, Synopsys is slightly undervalued, as we believe the market has yet to fully price in a successful Ansys merger approval.

Ban on EDA Software Sales to China Could Be Coming Eric Compton, CFA,Director,29 May 2025

Reports have emerged suggesting that the Trump administration is asking semiconductor software design firms, known as electronic design automation, or EDA, to cease selling to China. Why it matters: Sales to China made up 11% of revenue in the most recent quarter for Cadence. We see an increased likelihood that those sales may be at risk. While there is some uncertainty about what has happened, we have seen multiple reports about notices received by EDA firms, including previous reporting on Siemens. Therefore, we think it likely that EDA firms have or will receive some notice, even if

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descriptions of the details vary. This is despite management from Synopsys denying having received an official notice on their recent earnings call. While we don't think management is being untruthful, we believe this is likely an indication of the overall direction in which the administration is headed, regardless of whether a notice is given today. The bottom line: We think the current series of events increases the probability above 50% that EDA sales to China will be banned, and it is now our base case. We are reducing our fair value estimate for Cadence to $275 per share from $300. We have also reduced our fair value estimate for Synopsys. Our fair value estimate assumes that sales to China drop to $0 by the end of the year and that China is roughly margin neutral. We acknowledge that there is a high degree of uncertainty and our fair value estimate could change again if there is no ban or if the margin profile deviates from our expectations. On the bright side, we view our updated valuations as "de-risked" from China exposure. Our updated valuation for Synopsys is $510, and we still view the shares as moderately attractive. In contrast, we think shares of Cadence are slightly overvalued compared with our updated fair value estimate of $275.

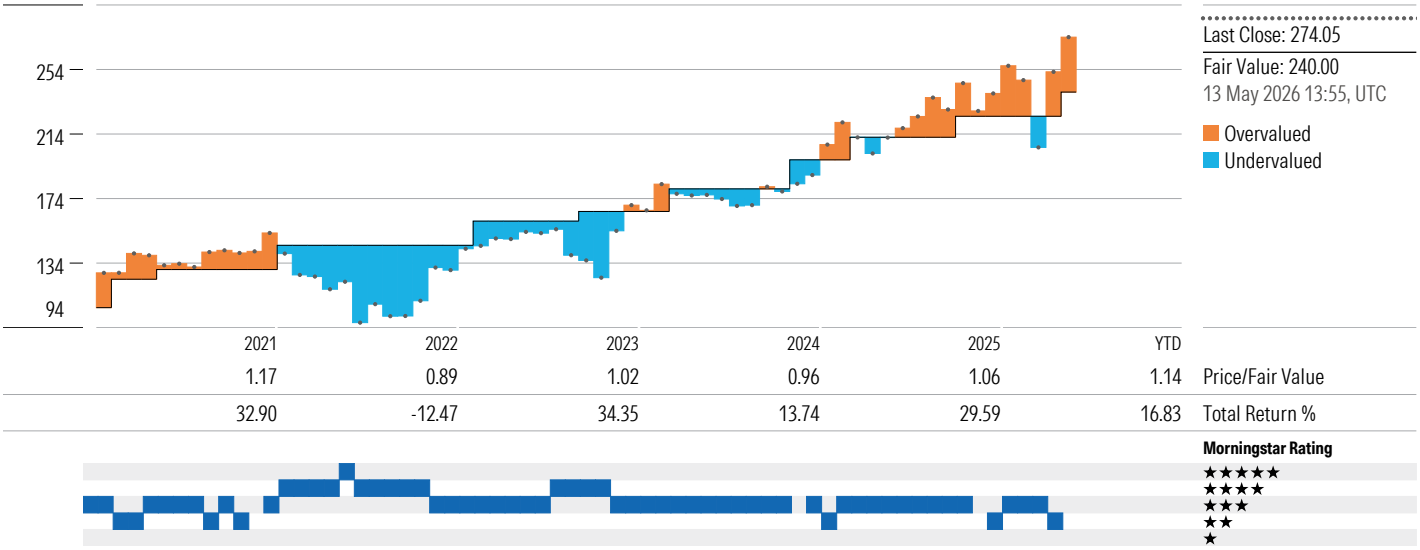
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Competitors Price vs. Fair Value

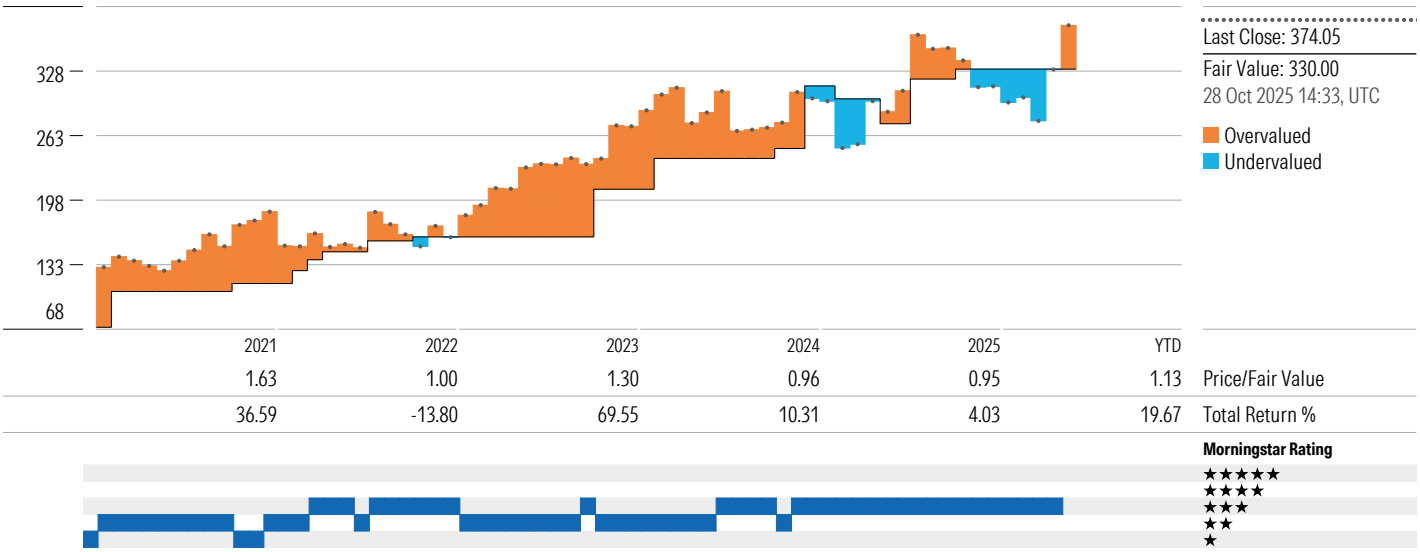
Siemens AG SIE



	2021	2022	2023	2024	2025	YTD	
	1.17	0.89	1.02	0.96	1.06	1.14	Price/Fair Value
	32.90	-12.47	34.35	13.74	29.59	16.83	Total Return %

Total Return % as of 27 May 2026. Last Close as of 27 May 2026. Fair Value as of 13 May 2026 13:55, UTC.

Cadence Design Systems Inc CDNS



	2021	2022	2023	2024	2025	YTD	
	1.63	1.00	1.30	0.96	0.95	1.13	Price/Fair Value
	36.59	-13.80	69.55	10.31	4.03	19.67	Total Return %

Total Return % as of 27 May 2026. Last Close as of 27 May 2026. Fair Value as of 28 Oct 2025 14:33, UTC.

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Morningstar Valuation Model Summary

Financials as of 27 May 2026

Fiscal Year, ends 31 Oct	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
Revenue (USD Mil)	5,843	6,127	7,054	9,703	10,619	11,703	12,924	14,238
Operating Income (USD Mil)	1,346	1,356	915	1,353	1,885	2,356	2,807	3,298
EBITDA (USD Mil)	1,516	1,651	1,575	2,912	3,654	4,157	4,661	5,218
Adjusted EBITDA (USD Mil)	2,196	2,553	2,789	2,912	3,654	4,157	4,661	5,218
Net Income (USD Mil)	1,230	2,263	1,332	520	1,265	1,727	2,202	2,648
Adjusted Net Income (USD Mil)	1,737	2,879	2,134	2,868	3,348	3,916	4,471	4,967
Free Cash Flow To The Firm (USD Mil)	808	1,235	-14,014	2,083	3,023	3,206	3,607	4,014
Weighted Average Diluted Shares Outstanding (Mil)	155	156	166	193	196	197	198	197
Earnings Per Share (Diluted) (USD)	7.92	14.51	8.04	2.70	6.46	8.78	11.13	13.45
Adjusted Earnings Per Share (Diluted) (USD)	11.19	18.46	12.88	14.87	17.10	19.90	22.61	25.22
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 27 May 2026

	3 Year Avg	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030	5 Year Avg
Operating Margin %	18.9	23.0	22.1	13.0	13.9	17.8	20.1	21.7	23.2	18.8
EBITDA Margin %	—	26.0	26.9	22.3	30.0	34.4	35.5	36.1	36.7	—
Adjusted EBITDA Margin %	—	37.6	41.7	39.5	30.0	34.4	35.5	36.1	36.7	34.5
Net Margin %	25.6	21.1	36.9	18.9	5.4	11.9	14.8	17.0	18.6	13.5
Adjusted Net Margin %	35.7	29.7	47.0	30.3	29.6	31.5	33.5	34.6	34.9	32.8
Free Cash Flow To The Firm Margin %	-54.9	13.8	20.2	-198.7	21.5	28.5	27.4	27.9	28.2	26.7

Growth & Ratios as of 27 May 2026

	3 Year CAGR	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030	5 Year CAGR
Revenue Growth %	11.6	15.0	4.9	15.1	37.6	9.4	10.2	10.4	10.2	15.1
Operating Income Growth %	-8.0	14.7	0.7	-32.5	47.8	39.4	25.0	19.1	17.5	29.2
EBITDA Growth %	4.5	9.1	8.9	-4.6	84.9	25.5	13.8	12.1	12.0	29.7
Adjusted EBITDA Growth %	15.6	21.5	16.2	9.2	4.4	25.5	13.8	12.1	12.0	13.4
Earnings Per Share Growth %	8.5	26.0	83.2	-44.6	-66.5	139.6	35.8	26.9	20.8	10.8
Adjusted Earnings Per Share Growth %	8.5	25.7	65.0	-30.2	15.4	15.0	16.4	13.6	11.6	10.8

Valuation as of 27 May 2026

	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
Price/Earning	42.0	27.8	35.2	35.4	30.8	26.4	23.3	20.9
Price/Sales	12.2	12.9	12.0	10.4	9.5	8.6	7.8	7.1
Price/Book	11.9	8.9	2.7	3.4	3.4	3.2	3.0	2.8
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	32.0	30.5	34.8	37.5	29.9	26.2	23.4	20.9
EV/EBIT	52.2	57.4	106.0	80.7	57.9	46.3	38.9	33.1
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 27 May 2026

Fiscal Year, ends 31 Oct	Actual 2023	2024	2025	Forecast 2026	2027	2028	2029	2030
ROA %	11.9	17.3	2.8	1.0	2.4	3.3	4.3	5.1
ROE %	19.9	25.1	4.7	1.8	4.1	5.3	6.4	7.2
ROIC %	18.6	16.0	-1.4	7.0	7.5	8.3	8.9	8.9

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Synopsys Inc SNPS ★★★ 28 May 2026 04:03, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
525.92 USD 27 May 2026	520.00 USD 28 May 2026 04:01, UTC	1.01	100.75 USD Bil 27 May 2026	Wide	Large Blend	High	Exemplary	27 May 2026 05:00, UTC

Financial Leverage (Reporting Currency)	Actual			Forecast				
Fiscal Year, ends 31 Oct	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Debt/Capital %	0.9	0.9	14.5	8.6	6.4	4.0	1.2	0.0
Assets/Equity	1.7	1.4	1.7	1.8	1.7	1.6	1.5	1.4
Net Debt/EBITDA	-0.6	-2.0	7.2	0.1	-0.7	-1.3	-2.0	-2.4
Total Debt/EBITDA	0.3	0.3	5.1	3.3	2.1	1.2	0.3	0.0
EBITDA/ Net Interest Expense	-67.5	-16.1	-5.8	6.3	10.6	20.8	81.5	-521.8

Forecast Revisions as of 27 May 2026	2026		2027		2028	
Prior data as of 25 Feb 2026	**Current**	**Prior**	**Current**	**Prior**	**Current**	**Prior**
Fair Value Estimate Change (Trading Currency)	520.00	514.54	—	—	—	—
Revenue (USD Mil)	9,703	9,623	10,619	10,531	11,703	11,605
Operating Income (USD Mil)	1,353	1,311	1,885	1,857	2,356	2,323
EBITDA (USD Mil)	2,912	2,872	3,654	3,632	4,157	4,130
Net Income (USD Mil)	2,868	2,832	3,348	3,322	3,916	3,883
Earnings Per Share (Diluted) (USD)	2.70	2.68	6.46	6.35	8.78	8.64
Adjusted Earnings Per Share (Diluted) (USD)	14.87	14.67	17.10	16.96	19.90	19.72
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 27 May 2026

Cost of Equity %	7.6
Pre-Tax Cost of Debt %	5.5
Weighted Average Cost of Capital %	7.6
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	9.6
Stage II Investment Rate %	25.0
Perpetuity Year	20

Additional estimates and scenarios available for download at https://pitchbook.com/.

Discounted Cash Flow Valuation as of 27 May 2026

	USD Mil
Present Value Stage I	27,747
Present Value Stage II	25,319
Present Value Stage III	53,283
Total Firm Value	**106,349**
Cash and Equivalents	2,888
Debt	13,485
Other Adjustments	0
Equity Value	**95,753**
Projected Diluted Shares	194
Fair Value per Share (USD)	**520.00**

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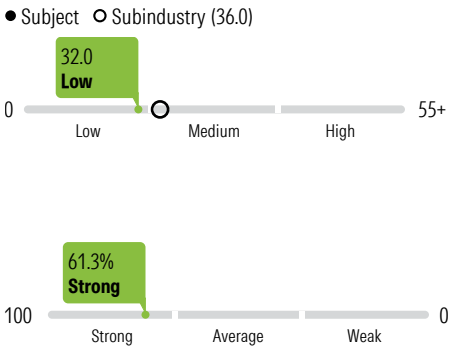
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Synopsys Inc SNPS ★★★ 28 May 2026 04:03, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
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ESG Risk Rating Breakdown

Exposure	
Company Exposure[1]	32.0
– Manageable Risk	29.9
Unmanageable Risk[2]	**2.1**
Management	
Manageable Risk	29.9
– Managed Risk[3]	18.3
Management Gap[4]	**11.6**
Overall Unmanaged Risk	**13.7**



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 61.3% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment[5]



ESG Risk Rating is of May 27, 2026. Highest Controversy Level is as of May 08, 2026. Sustainalytics Subindustry: Enterprise and Infrastructure Software. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 27 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure			Management			ESG Risk Rating		
Synopsys Inc	32.0 | Low	0	55+	61.3 | Strong	100	0	13.7 | Low	0	40+
Cadence Design Systems Inc	33.2 | Low	0	55+	60.2 | Strong	100	0	14.5 | Low	0	40+
Siemens AG	58.2 | High	0	55+	66.4 | Strong	100	0	23.1 | Medium	0	40+
Astera Labs Inc	31.9 | Low	0	55+	32.2 | Average	100	0	22.3 | Medium	0	40+
—	— | —	0	55+	— | —	100	0	— | —	0	40+

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Appendix

Historical Morningstar Rating

Synopsys Inc SNPS 27 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★★	★★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★	★★★	★★★	★★★	★★	★★	★★★	★★	★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★	★★★	★★★	★★★	★★	★★★	★★	★★★	★★★	★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★★	★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★

Siemens AG SIE 27 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★	★★★	★★★	★★★	★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
—	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
—	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
—	★★	★★★	★★	★★★	★★★	★★★	★★★	★★	★★	★★★	★★★

Cadence Design Systems Inc CDNS 27 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★	★★★	★★★	★★★	★★★	★★	★★	★★	★★	★★	★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★	★★	★★★	★★	★★	★★	★★	★★	★★	★★	★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★	★★★	★★★	★★★	★★	★★★	★★★	★★★	★★	★★	★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★	★	★★	★★	★★	★★	★★	★★	★★	★★	★★	★

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Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, indepth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss shortterm market-price movements), but we believe these negatives are mitigated by deep analysis and our longterm approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

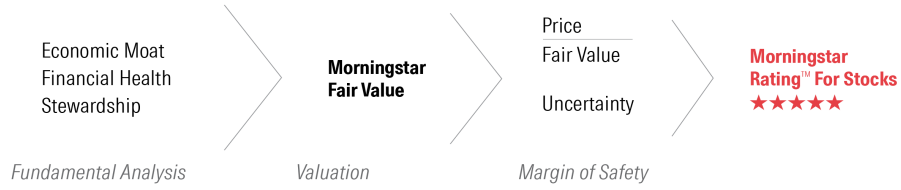
Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



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outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

	Margin of Safety	
Qualitative Analysis Uncertainty Ratings	★★★★★Rating	★Rating
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

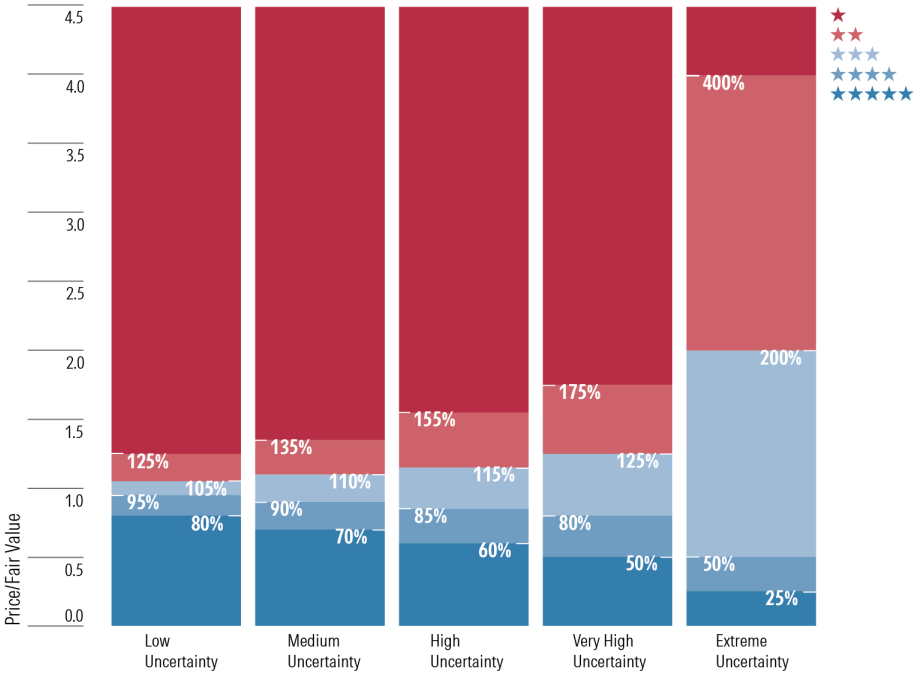
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to https://shareholders.morningstar.com

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multiyear time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

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